

Smart Medication Reminder System with ESP32 and Cloud Integration



SDG 11: Good Health and Well-Being

Presented by:-

S.Harini (312423106050)

S.Hema(312423106055)

Guided by:-

**Ms.S.Yuvashree.,M.Tech,
Assistant Professor.**

INTRODUCTION

- Maintaining proper medication schedules is important for elderly and chronic patients, but traditional reminder methods are unreliable and depend on user memory, leading to missed doses and poor medication adherence.
- Existing systems lack real-time monitoring, intake verification, and cloud connectivity, making healthcare management less effective.
- This project proposes a Smart Medication Reminder System using ESP32, IR sensor, buzzer, and AWS cloud integration for automated alerts, pill detection, and remote monitoring.
- The system improves medication adherence, reduces human error, and provides a reliable IoT-based healthcare solution.

ABSTRACT

- Smart Medication Reminder System is an IoT-based healthcare solution for improving medication adherence.
- The system uses ESP32, IR sensor, buzzer, and NTP for accurate reminders and pill detection.
- AWS cloud services enable real-time monitoring and data storage.
- The system reduces missed doses, improves patient safety, and supports smart healthcare management.

OBJECTIVE

- To provide timely medication reminders using ESP32 and buzzer alerts.
- To detect whether medicine is taken using an IR sensor.
- To reduce missed doses and human error in medication management.
- To enable real-time monitoring through AWS cloud services.
- To store medication history for tracking and analysis.
- To improve patient safety and medication adherence.

EXISTING SYSTEM

- Medication management is mainly done using manual reminders, alarm clocks, or mobile alarms.
- Existing systems depend heavily on user memory and caregiver support.
- Most systems only provide alerts without confirming medicine intake.
- No real-time monitoring or remote access is available.
- Medication records and history are not properly stored.
- Existing systems lack IoT and cloud integration for smart healthcare support.

PROPOSED SYSTEM

- The proposed system is an IoT-based Smart Medication Reminder System using ESP32.
- An IR sensor is used to detect whether the medicine has been taken.
- A buzzer provides alerts at scheduled medication times.
- NTP is used for accurate real-time scheduling.
- AWS IoT Core, Lambda, DynamoDB, CloudWatch,SNS enable cloud monitoring,data storage,alert.
- The system provides automated reminders, intake verification, and remote monitoring.
- It improves medication adherence and reduces human error.

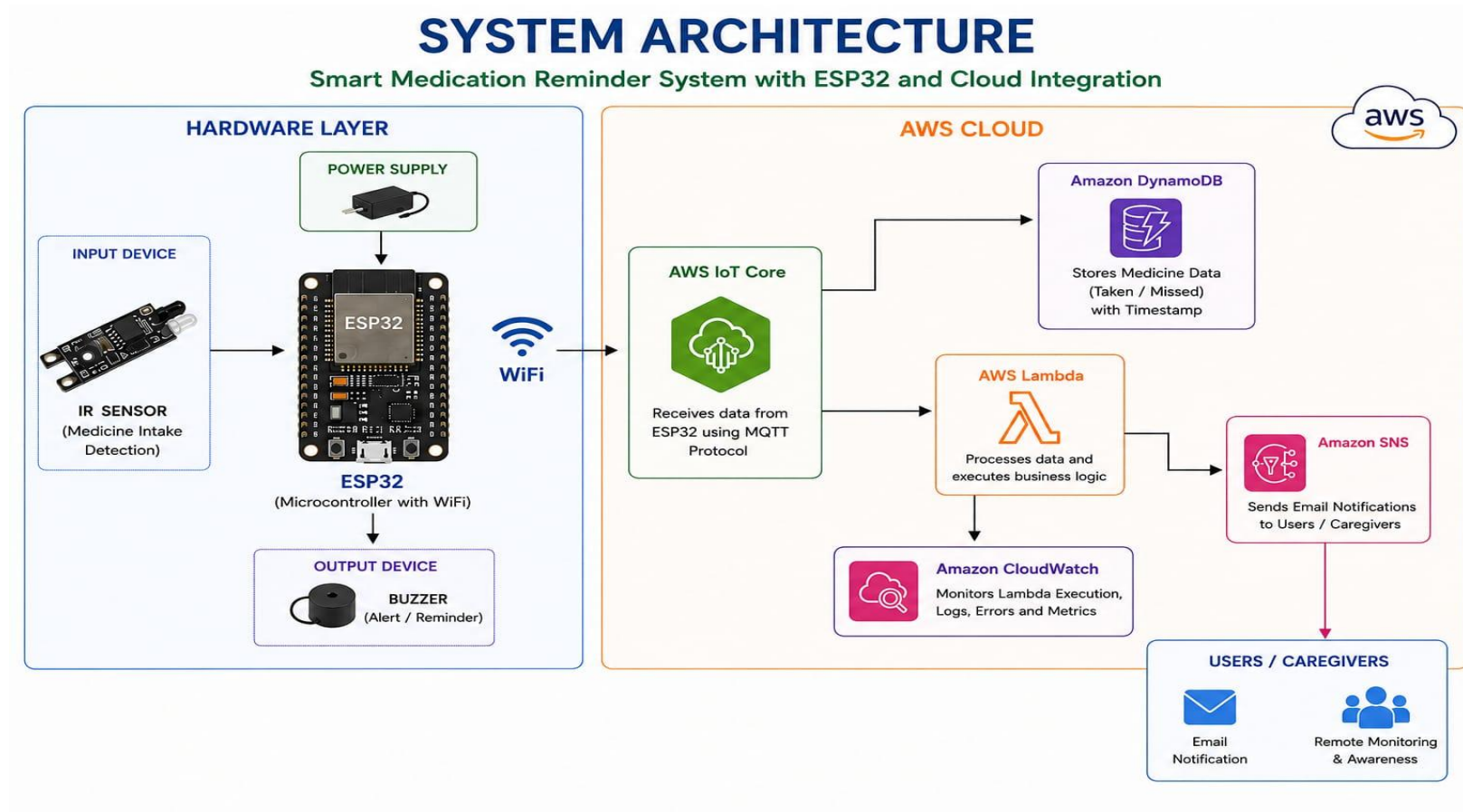
EXISTING SYSTEM vs PROPOSED SYSTEM

Aspect	Existing System	Proposed System
Method	Manual reminders / basic alarm systems	IoT-based system with ESP32 and cloud integration
Accuracy	Moderate, no intake confirmation	High, reliable intake verification using IR sensor
Speed	Slow and dependent on user action	Fast automated alerts and monitoring
Skill Requirement	Requires user memory or caregiver support	Minimal human intervention
Scalability	Limited functionality	Highly scalable with cloud support
Error Rate	High due to human error	Low with automated detection
Deployment	Local standalone setup	Cloud-based and remotely accessible
Maintenance	Manual monitoring and tracking	Automated cloud monitoring and updates

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER WITH AUTHOR NAME	JOURNAL NAME	YEAR OF PUBLICATION	METHODOLOGY	PROS & CONS
1.	Cloud-Connected Smart Medicine Dispenser Using ESP32 – R. Kumar and P. Sharma	International Journal of Advanced Research in Computer Science	2021	ESP32-based smart medicine dispenser with cloud communication	Pros: Real-time monitoring and cloud-based storage. Cons: Depends on internet connectivity and cloud availability.
2.	IoT-Based Pill Monitor and Reminder for Medication Intake – S. G. Jayandran and R. Mat Akir	Evolution in Electrical and Electronic Engineering	2022	ESP32-based smart pill reminder using IR sensor, buzzer, and Blynk cloud notification system	Pros: Real-time notifications and automated reminders. Cons: Limited scalability and depends on internet connectivity.

SYSTEM ARCHITECTURE



OUTPUT

Q Search mail

←

📁

🕒

🗑️

✉️

📧

⋮

1 of 109 < >

Medicine Reminder Alert Inbox x

AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

6:26 PM (0 minutes ago) ☆ 😊 ↶ ⋮

Medicine Slot 2 status: taken at 2026-05-13 12:56:18

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:470390348653:MedicineAlerts:f233392-88ba-4f55-977d-347eb9690e32&Endpoint=sevuganpriya98@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>

▶ 2026-05-13T12:56:18.397Z	START RequestId: c54938da-34fa-413a-9b4e-b1a28b8af42b Version: \$LATEST
▶ 2026-05-13T12:56:18.424Z	Received event: {"slot": 2, "status": "taken"}
▶ 2026-05-13T12:56:18.707Z	Data stored successfully
▶ 2026-05-13T12:56:18.956Z	Notification sent successfully
▶ 2026-05-13T12:56:18.958Z	END RequestId: c54938da-34fa-413a-9b4e-b1a28b8af42b
▶ 2026-05-13T12:56:18.958Z	REPORT RequestId: c54938da-34fa-413a-9b4e-b1a28b8af42b Duration: 559.91 ms Billed Duration: 560 ms Memory Size: 128 MB Max Memory Use..

CONCLUSION

- The Smart Medication Reminder System successfully provides automated medication management using IoT technology.
- ESP32, IR sensor, and buzzer ensure accurate reminders and pill intake verification.
- AWS cloud integration enables real-time monitoring and secure data storage.
- The system reduces missed doses, human error, and caregiver workload.
- It improves medication adherence, patient safety, and healthcare efficiency.
- The project provides a **reliable, scalable, and cost-effective** smart healthcare solution.

THANK YOU

